

Thesenpapier: Dual Spaces, Multilinear Forms, and Quadratic Spaces

Foundational Session 1 — Clifford Seminar

Clifford Seminar Group

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1 Foundational Definitions

1.1 Groups, Linear Spaces, and Algebras

Let G be a set equipped with an internal binary operation $\cdot : G \times G \rightarrow G$. The pair $(G, \cdot)$ constitutes a group if the operation satisfies the following three conditions:

$\mathcal{G}_1$ Associativity: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in G$.

$\mathcal{G}_2$ Identity element: there exists an element $e \in G$ such that $e \cdot a = a \cdot e = a$ for every $a \in G$.

$\mathcal{G}_3$ Inverse element: for each $a \in G$, there exists an element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

If $a \cdot b = b \cdot a$ holds for all $a, b \in G$, the group is abelian. An abelian group is written additively as $(V, +)$, with identity element 0 and inverse $-v$.

Let K be a field. A linear space over K is an additive abelian group $(V, +)$ equipped with an external scalar multiplication $\cdot : K \times V \rightarrow V$, denoted $(\lambda, v) \mapsto \lambda v$, satisfying the following conditions for all $\lambda, \mu \in K$ and all $u, v \in V$:

1. Distributivity over group addition: $\lambda(u + v) = \lambda u + \lambda v$.
2. Distributivity over field addition: $(\lambda + \mu)v = \lambda v + \mu v$.
3. Compatibility with field multiplication: $\lambda(\mu v) = (\lambda\mu)v$.
4. Unitary action: $1_K v = v$, where 1_K denotes the multiplicative identity of K .

The designation *vector space* is reserved strictly for concrete coordinate realizations such as $\mathbb{R}^n$ or explicit geometric displacement spaces.

Let V be a linear space over K . The structure $(A, +, \cdot, \bullet)$ is a K -algebra if $(A, +, \cdot)$ is a linear space over K equipped with a bilinear binary operation $\bullet : A \times A \rightarrow A$, satisfying for all $\lambda \in K$ and $u, v, w \in A$ the compatibility conditions

$$u \bullet (v + w) = u \bullet v + u \bullet w, \quad (u + v) \bullet w = u \bullet w + v \bullet w, \quad (1)$$

and

$$\lambda(u \bullet v) = (\lambda u) \bullet v = u \bullet (\lambda v). \quad (2)$$

The algebra is associative if $(u \bullet v) \bullet w = u \bullet (v \bullet w)$, and unital if there exists an element $1_A \in A$ such that $1_A \bullet u = u \bullet 1_A = u$ for all $u \in A$.

1.2 Dual Spaces and the Canonical Pairing

Let V be a linear space over K . The dual space V^* is the linear space of all linear functionals on V ,

$$V^* = \text{Hom}_K(V, K). \quad (3)$$

The evaluation of a functional $\alpha \in V^*$ on an element $v \in V$ is expressed by the bilinear dual pairing

$$\langle \cdot, \cdot \rangle : V^* \times V \rightarrow K, \quad \langle \alpha, v \rangle = \alpha(v). \quad (4)$$

The double dual space is $V^{**} = \text{Hom}_K(V^*, K)$. The canonical evaluation mapping $\text{ev} : V \rightarrow V^{**}$ defined by

$$\text{ev}_v(\alpha) = \langle \alpha, v \rangle \quad (5)$$

is an injective linear homomorphism. When $\dim V < \infty$, the map ev is an isomorphism. Consequently, elements of V act natively as linear functionals on V^* without auxiliary structure.

1.3 The Free Tensor Algebra

The tensor algebra of a linear space V over K is the graded associative K -algebra

$$T(V) = \bigoplus_{r=0}^{\infty} T^r(V), \quad (6)$$

where $T^0(V) = K$, $T^1(V) = V$, and $T^r(V) = \bigotimes^r V$ for $r \geq 2$. Multiplication is governed by the associative tensor product $\otimes$. The structure $T(V)$ satisfies a universal property: every linear map from V into an associative unital K -algebra A extends uniquely to a unital algebra homomorphism from $T(V)$ into A . The tensor algebra imposes no algebraic relations between distinct elements of V .

1.4 Quadratic Forms and Polarization

Let V be a real linear space. A map $q : V \rightarrow \mathbb{R}$ is a quadratic form if it satisfies $q(\alpha v) = \alpha^2 q(v)$ for all $\alpha \in \mathbb{R}$ and $v \in V$, and the associated symmetric bilinear form $b : V \times V \rightarrow \mathbb{R}$ defined by the polarization identity

$$b(u, v) = \frac{1}{2}(q(u+v) - q(u) - q(v)) \quad (7)$$

is linear in each argument. The radical of the quadratic space (V, q) is the subspace

$$\text{rad}(V) = \{u \in V \mid b(u, v) = 0 \text{ for all } v \in V\}. \quad (8)$$

The space (V, q) is non-degenerate if $\text{rad}(V) = \{0\}$.

A symmetric bilinear form induces a linear mapping $b^\flat : V \rightarrow V^*$ given by $b^\flat(u) = b(u, \cdot)$. Non-degeneracy of q is equivalent to $b^\flat$ being an isomorphism, whose inverse is denoted $b^\sharp : V^* \rightarrow V$.

By Sylvester's law of inertia, any finite-dimensional real quadratic space decomposes into an orthogonal direct sum $V = V_+ \oplus V_- \oplus V_0$ such that $q|_{V_+}$ is positive definite, $q|_{V_-}$ is negative definite, and $q|_{V_0} = 0$. In the non-degenerate case, $\dim V_0 = 0$, and the signature is denoted (p, m) , where $p = \dim V_+$ and $m = \dim V_-$. The resulting pseudo-Euclidean vector space is denoted $\mathbb{R}^{p,m}$.

2 Physical Foundations: Kinematics versus Measurement

In classical Lagrangian mechanics on a configuration manifold M , generalized velocities $\dot{x}$ belong to the tangent space $T_x M$, which is a linear space of directional derivations. In contrast, generalized forces and canonical momenta $p_i = \partial_{\dot{x}^i} L$ reside in the cotangent space $T_x^* M$, which is the dual linear space. The pairing $\langle p, \dot{x} \rangle$ represents the mechanical work rate and requires no metric structure.

The conventional identification of velocities with momenta in Newtonian mechanics relies on the kinetic metric $b(u, v) = m \sum_i u^i v^i$, which supplies the isomorphism $b^\flat : T_x M \rightarrow T_x^* M$. Kinetic energy $T = \frac{1}{2} q(\dot{x})$ is not a linear invariant, but the evaluation of this specific quadratic form.

In relativistic mechanics, the causal structure of spacetime is established prior to the introduction of coordinates or curved geometry. Given the real linear space V of spacetime displacements, the choice of a non-degenerate quadratic form of signature $(1, 3)$ divides V into timelike ($q(v) > 0$), spacelike ($q(v) < 0$), and lightlike ($q(v) = 0$) elements. Light cones are the algebraic zero-sets of q , demonstrating that relativistic causality is an algebraic property of quadratic spaces.

3 Derivation Agenda for the Blackboard Exposition

The 40-minute Referat follows four sequential stages:

1. Axiomatic synthesis of linear spaces from additive abelian groups and external field actions, followed by the coordinate-free definition of V^* and the canonical isomorphism $V \cong V^{**}$.
2. Construction of the free tensor algebra $T(V)$ as the maximal multilinear structure, demonstrating its infinite-dimensional, relation-free nature.
3. Introduction of the quadratic form q , derivation of the symmetric bilinear form via the polarization identity, and the construction of the musical isomorphisms $b^\flat$ and $b^\sharp$.
4. Orthogonal decomposition of non-degenerate quadratic spaces, proof of signature invariance, and classification of isotropic rays $q(v) = 0$.

4 Theses for Plenary Discussion

Thesis I The standard textbook practice of identifying velocities with canonical momenta via Cartesian index identification ($p_i = m\dot{x}_i$) obscures the duality between V and V^* . This obscures the geometric foundation of Hamiltonian mechanics and creates conceptual barriers when transitioning to curved manifolds or relativistic field theories.

Thesis II The free tensor algebra $T(V)$ contains excess degrees of freedom that lack geometric meaning. A physical geometry cannot emerge from multilinear algebra until an algebraic ideal generated by measurement—specifically the quadratic form $q(v)$ —is factored out.

Thesis III A metric is not an intrinsic property of a linear space, but an auxiliary constitutive mapping $b^\flat : V \rightarrow V^*$ that converts displacements into linear measurement functionals.

Thesis IV Causal structure in relativistic physics is an algebraic consequence of the signature $(1, 3)$ on a four-dimensional linear space, independent of differential geometry or pseudo-Riemannian metric tensors.

5 Preparatory Exercises

1. Consider the set $\mathcal{U}$ of coherent physical units generated by the base units of the International System of Units (SI). Let group addition on $\mathcal{U}$ be defined by physical unit multiplication (for example, $\text{m} \cdot \text{s} = \text{m s}$ and $\text{m} \cdot \text{m} = \text{m}^2$), and define external scalar multiplication by $\mathbb{Q}$ as unit exponentiation, $(\lambda, u) \mapsto u^\lambda$. Show that $(\mathcal{U}, \cdot)$ satisfies the axioms of an additive abelian group, with the dimensionless unit 1 serving as identity, and verify that this structure satisfies all axioms of a linear space over $\mathbb{Q}$.
2. Let the ordered set of seven base units $\mathcal{B} = \{\text{m, kg, s, A, K, mol, cd}\}$ form a basis for the SI linear space over $\mathbb{Q}$. Determine the explicit coordinate tuples in $\mathbb{Q}^7$ corresponding to:
 - (a) The Newton (N), unit of mechanical force.
 - (b) The Farad (F), unit of electrical capacitance.
 - (c) The Lumen (lm), unit of luminous flux.
3. Let V be a finite-dimensional linear space over K , and let $\mathcal{E} = \{e_i\}_{i=1}^n$ be a basis for V . The dual basis $\mathcal{E}^* = \{\omega^j\}_{j=1}^n \subset V^*$ is defined by $\langle \omega^j, e_i \rangle = \delta_i^j$. Show that the mapping $e_i \mapsto \omega^i$ depends explicitly on the choice of basis, confirming that no canonical isomorphism between V and V^* exists. In contrast, prove that the evaluation map $\text{ev} : V \rightarrow V^{**}$ defined by $\text{ev}_v(\alpha) = \langle \alpha, v \rangle$ is completely basis-independent and canonical.
4. In classical mechanics, consider a free particle on $\mathbb{R}^3$ described by the Lagrangian $L(v) = \frac{1}{2}mb(v, v)$, where b is a symmetric, positive-definite bilinear form. Show that the definition of canonical momentum $p = \partial_v L$ is identical to the action of the musical isomorphism $b^\flat : V \rightarrow V^*$. Compute $b^\sharp : V^* \rightarrow V$ and show that the classical Hamiltonian $H(p) = \langle p, v \rangle - L(v)$ corresponds to the inverse quadratic form on the dual space.
5. Consider the real linear space $V = \mathbb{R}^4$ equipped with the Minkowski quadratic form $q(x) = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2$.
 - (a) Apply the polarization identity to determine the explicit symmetric bilinear form $b(x, y)$.
 - (b) Show that $\text{rad}(V) = \{0\}$, establishing non-degeneracy.
 - (c) A linear subspace $W \subset V$ is totally isotropic if $q|_W = 0$. Prove that the maximum dimension of any totally isotropic subspace in $\mathbb{R}^{1,3}$ is 1, and explain why physical light rays cannot form a two-dimensional planar subspace.

6 Preparatory Literature

Participants are expected to study the following references prior to the session:

- Abstract groups and linear spaces: Fraleigh [1, Chaps. 1.1, 1.2, and 7.33]; Axler [2, Chaps. 1 and 2].
- Dual spaces, multilinear forms, and quadratic spaces: Vaz and da Rocha [3, Chap. 1].

References

- [1] John B. Fraleigh and Victor J. Katz. *A First Course in Abstract Algebra*. 8th ed. World Student Series. Hoboken, New Jersey: Pearson, 2021. 424 pp. ISBN: 978-0-13-575816-8.

- [2] Sheldon Axler. *Linear Algebra Done Right*. Undergraduate Texts in Mathematics. Cham: Springer International Publishing, 2015. ISBN: 978-3-319-11079-0. DOI: 10.1007/978-3-319-11080-6. URL: <https://link.springer.com/10.1007/978-3-319-11080-6>.
- [3] Jayme Vaz and Roldão da Rocha. *Introduction to Clifford Algebras and Spinors*. Oxford: Oxford university press, 2016. 128 pp. ISBN: 978-0-19-878292-6.